

**CSCI\_6461**  
**Computer System Architecture**

**Team 8**  
**14<sup>th</sup> April, 2025**

**Project Part-3**

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**User Manual**

## How to run the Simulator?

1. Download the jar file named – “Team 8 Simulator part 3 CSCI 6461\_12”.
2. Download Program1.txt, Program2.txt and Demo.txt and keep in mind where you downloaded it.

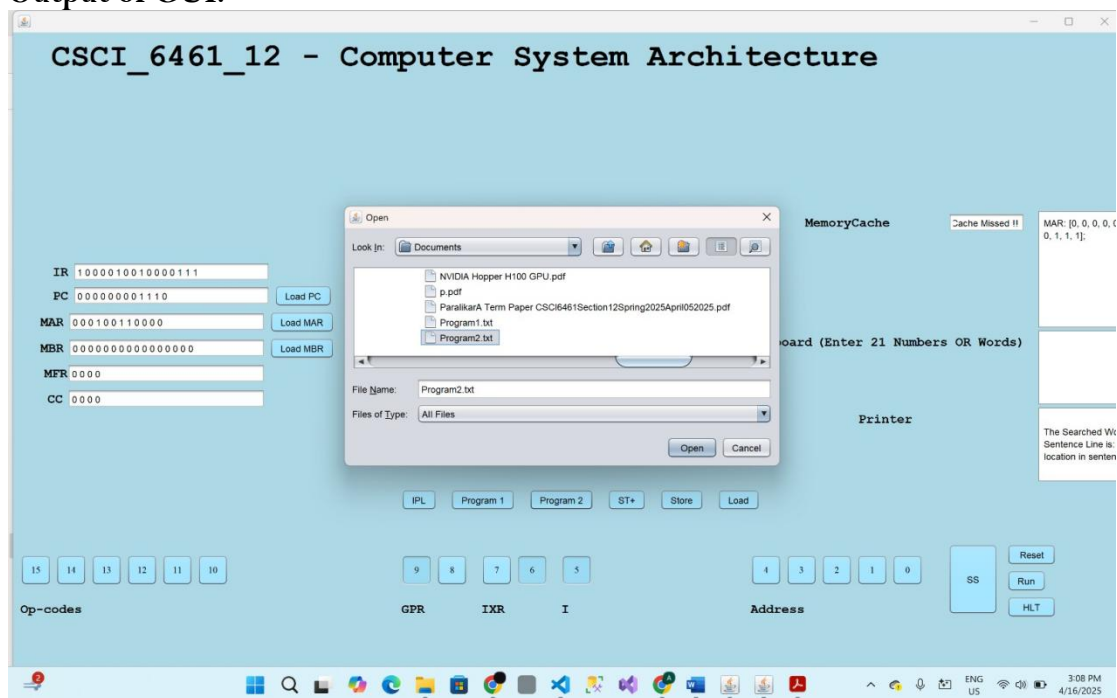
Double-clicking the downloaded jar file will launch it immediately; to test it, click the IPL button and upload the load.txt file (the steps to do this are mentioned below in the Testing the Simulator section).

3. Enter `java -jar sim.jar` in the terminal after `cd` into the location where you got the jar file.

**Open the source code in any IDE, navigate to the GUI.java file and launch the main application there if you choose to run it straight from the source code.**

## Testing the Simulator

### Output of GUI:



**Once the simulator is up and running, you can start testing the simulator.**

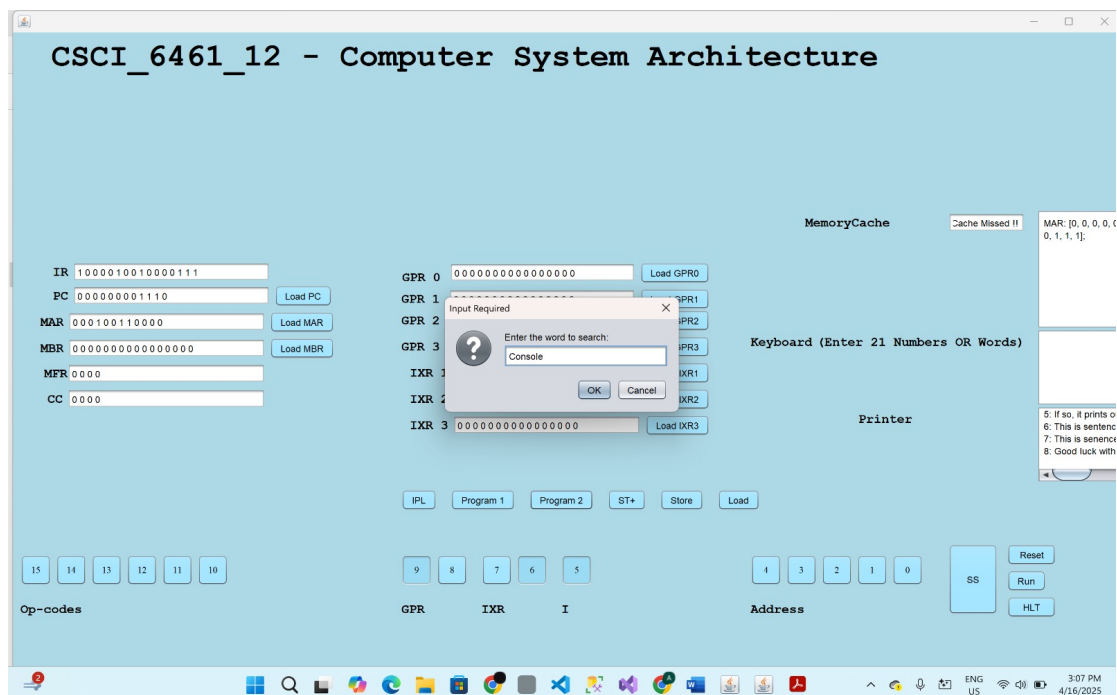
1. On the user interface, click the program2 button. A dialog window will appear as seen in the image above.
2. Choose the Program2.txt file after navigating to the directory. We're presuming that an assembler has already analyzed the code.
3. You will receive a notification in the console after uploading it, indicating that the file upload was successful.

```

1  import java.awt.event.*;
2  import java.io.*;
3  import java.io.IOException;
4  import javax.swing.*;
5  import java.util.*;
6  import java.util.Scanner;
7
8  public class GUI extends JFrame {
9
10     // Global variables
11     CacheMemory MemoryCache = new CacheMemory(16);
12     InstructionFormatter strFormatter = new InstructionFormatter();
13     int[] instructionArray = new int[] { 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0 };
14     int[] default_PC_loc = new int[] { 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0 };
15     boolean run_check = false;
16
17     int numberCount = 0;
18     private static final int MAX_NUMBERS = 20;
19     private static int[] enteredNumbers = new int[MAX_NUMBERS];
20
21     // Function that runs the GUI and Processor
22     public GUI() {
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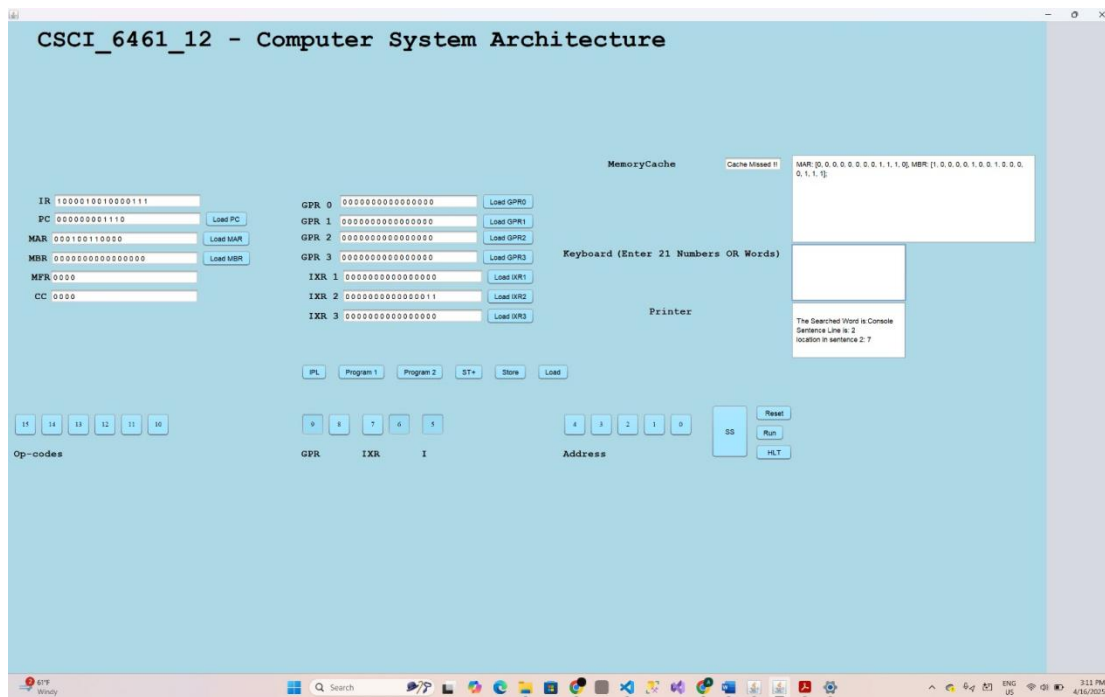
```

- The simulator must now be tested. Observe how the value in PC changes on its own. It shifts to the location of the loading file's first instruction. You will be prompted to search for the term in the loaded file by a popup.



- Enter any keyword from the file to get started. In the event that we search for "word console," for instance, the printer will display both the word number and the sentence number where the word appears.

6. The output will be shown as below:



## Terminal:

```
Setting Memory for Row 303
Setting Memory for Row 304
Loaded Content in Demo File is :
1: A program that reads a set of a paragraph of 6 sentences from a file into memory.
2: It prints the sentences on the console printer.
3: It then asks the user for a word.
4: It searches the paragraph to see if it contains the word.
5: If so, it prints out the word, the sentence number, and the word number in the sentence.
6: This is sentence five.
7: This is sentence six.
8: Good luck with testing.
The Searched word is: "Console"

Sentence Line is: 2
Location in sentence 2 : 7
█
```